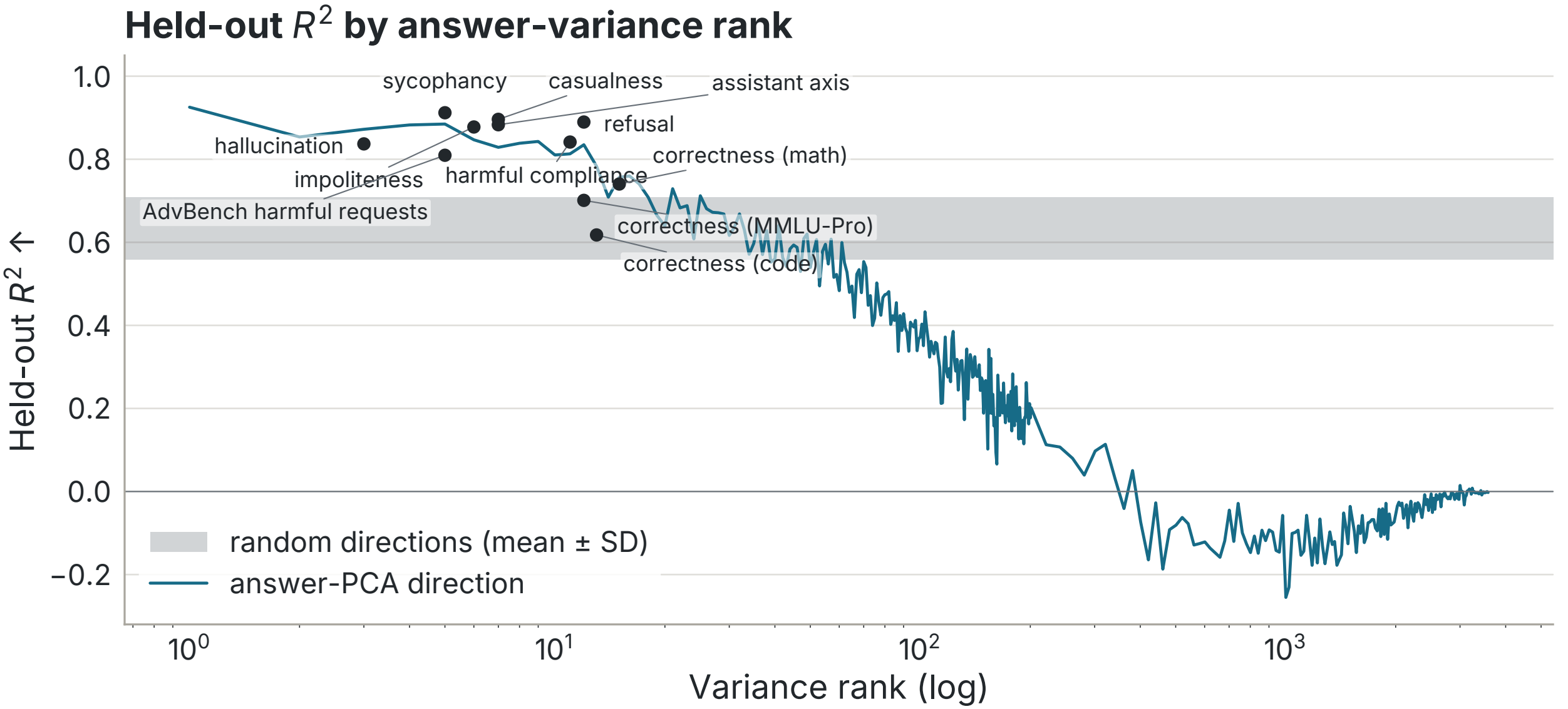
Held-out R^2 by answer-variance rank
sycophancy
casualness
assistant axis
refusal
hallucination
correctness (math)
impoliteness
harmful compliance
AdvBench harmful requests
correctness (MMLU-Pro)
correctness (code)
random directions (mean ± SD)
answer-PCA direction
Held-out R^2 ↑
Variance rank (log)
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
10^0
10^1
10^2
10^3